

No vertebrate has lost an IP₃ receptor paralogue

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Abstract

Vertebrates carry three inositol 1,4,5-trisphosphate receptor genes, *ITPR1*, *ITPR2* and *ITPR3*, and gene families of that age normally lose copies in some lineages. Whether any vertebrate has lost one of the three has never been asked of the genomes themselves, and a survey that reports no loss is only as good as its measured ability to find a gene that is there. We aligned a 38-protein bait panel, with the ryanodine receptors as an internal positive control, to a declared scope of 309 vertebrate assemblies and recorded 1,236 genome × cell results. The control fired in every genome. We then restated all 927 genome × paralogue cells under eight ordered positive tests, of which exactly one licenses a loss. No cell reaches it, and Dollo parsimony places zero losses on the family character and on each paralogue. We measured the search's own miss rate on the same data: because no loss is reconstructed, every cell whose gene is independently known to be present and was not found is a false negative, 140 of 923 cells overall, falling to 5 of 563 in assemblies whose contigs are longer than the gene, with an independent ryanodine receptor series agreeing. An exact ablation of the bait panel shows that paralogue coverage, not phylogenetic breadth, sets recall. Finally, we name what it would take to manufacture a loss: the family-level coding yields one in 2 of 32 analytical settings, the paralogue-resolved coding in 18 of 32. Every vertebrate genome able to hold the gene carries all three paralogues.

Introduction

Inositol 1,4,5-trisphosphate receptors (IP₃Rs) are the ligand-gated calcium release channels of the endoplasmic reticulum, the step that turns a receptor-generated second messenger into a cytosolic calcium signal [1, 2]. Vertebrates carry three paralogues, *ITPR1*, *ITPR2* and *ITPR3*, which arose in the two whole-genome duplications at the base of the vertebrates (companion paper 2) and differ in tissue distribution and regulation [3, 4]. Their individual loss in mice has distinct consequences: ataxia and seizures without type 1 [5], and failure of exocrine secretion when types 2 and 3 are removed together [6]. In humans, deletions and point variants of *ITPR1* cause spinocerebellar ataxia [7, 8, 9], and loss of type 2 function abolishes sweating [10].

That each paralogue matters in one mammal does not say that each is indispensable across the vertebrates. Gene loss is a common outcome of duplication, and families as old as this one usually show it somewhere [11]. A lineage that had dropped one IP₃R paralogue would be a natural experiment in calcium signalling, and finding one requires only a genome search. The difficulty is not the search but the negative. A survey that reports that no lineage has lost a gene is making a claim about everything it failed to find, and its value depends on how often

the same search misses a gene that is really there. Genome assemblies are uneven across the vertebrates [12], annotation is less reliable still [13], and an IP₃R gene spans 76 kb to half a megabase of genomic sequence, so a fragmented assembly can hide one without any biological cause.

Here we ask whether any vertebrate lineage has lost one of its three IP₃ receptor genes, and we built the analysis so that the answer could be a measurement in either direction. The scope is 309 vertebrate assemblies, declared before the search. Each is searched by spliced protein-to-genome alignment [14] with a bait panel that carries the ryanodine receptors, the IP₃R sister family, as an internal positive control. Every genome × paralogue cell is then restated under a vocabulary in which exactly one state can be counted as a loss, reachable only after every alternative explanation has been excluded by a positive test. Cells the aligner could not decide are resolved by reassembling the reference across contigs, against a bar calibrated on a gene known to be elsewhere.

Two measurements make the result readable. The first is the search's own false-negative rate, obtained from the same data: because the count turns out to be zero, every cell whose gene is independently known to be present and which the search did not find is a miss with a known right answer. The ryanodine receptor cell, present in every vertebrate and scored by none of the same rules, provides a second, independent series. The second is a sensitivity matrix that names the analytical settings under which a loss would appear, so that the zero is reported with its escape routes rather than as a single number.

Results

The sweep searched every genome in a declared scope with a control that never failed

The scope is 309 vertebrate assemblies, 552,535,418,825 bp in total: one best assembly for each of 161 vertebrate orders, and 169 margin species that the protein record left in doubt (a species that is both is one row). The margin species were derived from the protein census by stated rules rather than listed by hand: their reference proteome returned no family record, carried fewer than three paralogues, held only fragments, or was an anchor species. They are therefore enriched, by construction, for genomes in which a paralogue looks missing, which is the population a loss survey most needs and the one in which a false absence is most likely.

A panel of 38 baits (30 IP₃ receptors and 8 ryanodine receptors, 121,294 residues) was aligned to every genome, and cells the alignment left empty were re-asked with a whole-genome translated search [15]. All 309 genomes completed. The ryanodine receptor control, which should fire in every vertebrate, failed in 0 of them, so no genome is excluded on control grounds. The sweep recorded 1,236 genome × cell results over 2,144 loci (Fig. 1; Extended Data Fig. 1). At every one of the 2,144 loci only one family's baits aligned, so the family question is settled by the alignment before any threshold is applied.

Read as a raw ledger, the result is already almost complete. Four cells in 1,236 were called absent, meaning no aligned locus and no remnant, and all 4 are in the two cyclostomes, *Petromyzon marinus* and *Myxine glutinosa*, which lack *ITPR2* and *ITPR3* in the ledger. The rest of the missing evidence is not absence but weaker presence: truncated loci, fragments and translated-search traces, concentrated in particular vertebrate classes (Fig. 1). The remainder of this paper asks whether any of that is a loss.

The search misses a gene only where the assembly is fragmented

The central control of any absence claim is how often the same search fails to find a gene that is present. We measured it on the search's own output. Anticipating the result below, that no loss is reconstructed anywhere in the scope, every cell whose gene is independently known to be present and which the ledger did not grade as found is a false negative of the method, a miss with a known answer. Over the whole scope that is 140 of 923 cells, a rate of 0.1517 (Fig. 2a).

That series depends on the state rules of the next sections, so we carried a second one that does not. Every

vertebrate has ryanodine receptors, so every genome's control cell has a known answer that no part of the state logic produced. It misses 42 of 309, a rate of 0.1359, indistinguishable from the IP₃R series (Fisher $p = 0.578$). The miss rate is a property of the method applied to these assemblies, not of the paralogue calls.

Every miss tracks assembly contiguity. A missed cell's assembly has a median contig N50 of 23,460 bp against 3,396,515 bp for a found one; the odds of finding the gene rise 8.10-fold per tenfold increase in contig N50 in the IP₃R series and 19.99-fold in the ryanodine series; and chromosome-level assemblies miss 3 of 512 IP₃R cells and 0 of 172 ryanodine cells (Fig. 2a; Extended Data Fig. 2). Below the contiguity bar the miss rate is 0.3917 for *ITPR1*, 0.4333 for *ITPR2* and 0.3 for *ITPR3*.

The contiguity bar was fixed before any error rate was known: 142,212 bp of contig N50, the median measured genomic span of the gene, on the grounds that an assembly whose contigs are shorter than the gene cannot represent it on one contig. Scored against the miss rate afterwards, it holds. Above it, 189 genomes carry 563 IP₃R control cells, of which 5 are missed (a rate of 0.0089), and the ryanodine series misses none of 189 (Fig. 2b). The bar has a price, which is stated rather than absorbed: it sets aside 120 of 309 genomes, disproportionately birds and margin species (Fig. 2c). Every count below is therefore given over all 309 genomes and over the 189 above the bar.

The bait panel's recall depends on paralogue coverage, not on phylogenetic breadth

The spliced aligner treats each bait independently, so dropping baits from the retained alignments and re-running the cell-assignment chain reproduces exactly what the sweep would have reported with a smaller panel. The simulation reproduces the committed ledger in 1,236 of 1,236 cells, with 0 mismatches, so each of 19 panels, the full one and 18 ablations, is measured rather than modelled (Fig. 3).

The full panel recovers 783 cells. Four human baits, one per cell, recover 782 of them, and removing any single vertebrate clade band changes the count by at most two cells in either direction. Removing one paralogue's own baits is what costs recall: the deltas are -238, -240 and -241 cells for *ITPR1*, *ITPR2* and *ITPR3* (Fig. 3a). A single bait recovers its gene at any identity above 0.5 (Fig. 3b). Removing the ryanodine receptor baits changes no IP₃R call, so the positive family test costs nothing in recall. Of the 2,179 call changes across all panels, 68 are gains, each produced by a marginal cell whose coverage is read from its top-scoring bait and which a competitor's removal pushed across the coverage bar. They are properties of the assignment rule, not evidence that a smaller panel searches better.

No genome × paralogue cell reaches the state that licenses a loss

The ledger's own statuses were not designed to license a loss count. There, "absent" means that the rescue search attributed no region, which in a shattered assembly is a statement about contig lengths. We therefore restated each of the 927 genome × paralogue cells under eight ordered rules, in which the first rule to fire assigns the state and exactly one state is countable as a loss. A cell is uninformative if its genome's control failed. It is present if it carries a locus at or above the coverage bar; truncated if its locus ends at a contig edge or a run of ambiguous bases; partial if its locus is short with no assembly excuse; fragmented if the reference reassembles from sequence outside every locus the aligner placed; and paralogue-unassignable if the genome carries family loci that no cell claimed. Only then can a cell be undecidable because its assembly cannot hold the gene on one contig, or absent. The absent state is reachable by construction: a self-test builds a contiguous, controlled, empty, spare-free cell and requires it to come back absent.

Of the 927 cells, 783 are a single full-coverage locus, 90 a locus truncated by the assembly, 7 a partial locus, 43 a gene reassembled across contigs and 4 paralogue-unassignable. None is absent, and 923 hold a gene that is present (Fig. 4a).

The four unassignable cells are the ledger's four absences. Both cyclostome genomes carry three full-length family loci, all of which fall to the *ITPR1* baits because the panel holds no cyclostome-labelled *ITPR2* or *ITPR3* bait. Filing those genomes' *ITPR2* and *ITPR3* cells as absent would score two losses per cyclostome that the genomes themselves contradict. They are counted neither as present nor as lost.

Genomic sequence, unlike reference coverage, can count copies, because a placed locus and a reassembly occupy different places in the genome. Summed per genome, every one of the 189 assemblies above the contiguity bar holds a minimum of 3.0 gene-equivalents, all three paralogues. Below the bar, 11 of 120 fall short of 2.5, and the shortfall follows contig N50 rather than taxonomy (Fig. 4b).

Every undecided cell reassembles into one gene above a calibrated bar

The 43 fragmented cells are the ones a naive count would read as candidate losses: the aligner placed no locus, but the translated search found something. For each we measured the non-redundant coverage of one reference protein reassembled across contigs, computed only from sequence outside every locus the aligner clustered in that genome. Both restrictions are necessary in a family whose paralogues are 61 to 68 % identical: without the exclusion set, a missing gene reassembles out of its own paralogues, and a union over several baits reports a coverage that no single protein achieves.

The bar a reassembly must clear was measured against a gene known to be elsewhere. The negative population is the set of regions the full panel attributes to a paralogue whose gene the aligner already placed at a locus in the same genome, so the fragments cannot be that gene and measure only what cross-paralogue similarity delivers on its own. The 44 positive regions reach a median coverage of 0.7948 and the 9 negative ones 0.0303 (Fig. 5a). The first version of this calibration did not separate, and the reason is itself a result: in a genome where two paralogues are both shattered, a fragment attributed to the other one is a piece of a real gene. Those 20 co-shattered regions, at a median of 0.6308, are now reported as their own population, the measured size of the paralogue-attribution problem in a fragmented assembly. With them separated, Youden's index [16] is 1 and the gap runs from 0.1 to 0.1737. The operating point is the gap's midpoint, 0.13685, rather than Youden's own threshold, which on a perfectly separated pair is the most permissive bar the data allow. Every undecided cell clears the bar, whether its gene is spread over two contigs or twelve (Fig. 5b).

We also asked whether flanking-gene synteny could independently place these fragments, using a neighbourhood caller that is correct at every key count at which it acts. It could not reach them. Of 432 rescue regions, 273 sit on a contig that carries no annotated gene at all and 8 reach the caller's four-key floor; the median region's contig extends 39.936 kb, shorter than the gene (Extended Data Fig. 3). Synteny is accurate here and unavailable, a property of the assemblies rather than of the method.

Parsimony places no loss, and the settings that would manufacture one are named

Under the operating point, on the family-level coding and on each paralogue, the character is present in every genome that can be scored, and Dollo parsimony [17] over the NCBI-taxonomy tree of all 309 assemblies places 0 losses. The count is zero because no cell reaches the loss state, not because the routine cannot return anything else: constructed controls put a loss on a known edge and require it found, and require two sister losses merged into their parent. The tree is an input, not a result, and its 468 internal nodes include 49 polytomies, which bound any count of independent losses from above but cannot create one where there is none.

With no loss to place, the informative quantity is how far the rules must move before a loss appears. We walked 32 settings: an ordered eight-rung evidence ladder, each rung named after what it refuses, crossed with the cyclostome rule and the contiguity rule each switched on and off, on both codings and under three branch-length schemes (Fig. 6). The first three rungs are the calibration's own gap edges and midpoint, so the first

question is whether moving the reconstruction bar across its measured uncertainty changes anything; it changes no cell in 927 (Extended Data Fig. 4).

The two codings then behave differently. The family-level coding manufactures a loss in 2 of 32 settings, at worst 1 genome in 309, and only when everything except a complete locus is refused and the contiguity bar is ignored at the same time (Fig. 6a). The paralogue-resolved coding manufactures one in 18 of 32 settings, up to 45 loss edges (Fig. 6b). A per-paralogue absence is fragile to every one of these settings; a family-level absence is not. Branch lengths change the count in 0 of the 32 settings, as parsimony requires.

Continuous-time Markov models of the character [18] are stated and not fitted. An invariant character contains no transition to estimate, and at all twelve model, axis and branch-length combinations the likelihood profile is monotone to the edge of an eight-order-of-magnitude rate grid (Extended Data Fig. 5). No rate is reported for it.

No reading frame marks a dead gene

Frameshifts and premature stops counted along each placed locus would, read naively, be a pseudogene screen. We counted them per kilo-aligned-residue and found that the confounder is not the obvious one: assembly contiguity barely moves the lesion count, whereas a locus's identity to its bait moves it a great deal (Extended Data Fig. 6), because a distant reference buys its alignment with frameshifts. Compared within each genome against the same genome's other family loci and matched on identity, one paralogue shows an excess: *ITPR3* carries more lesions than its identity-matched siblings in 39 genomes to 14, while the unmatched *ITPR2* excess disappears once identity is matched.

That excess is not decay. Of 1,760 scored loci, 44 clear the lesion bar, and all 44 are full-coverage loci with an intact gene model; 7 family loci fire any of three deliberately generous fossil readings, each on one or two stops in a complete model. There are therefore no dead loci on which a shared-lesion test of an ancestral pseudogene could be run. Stratified by class, the *ITPR3* excess is a bird result, 25 genomes to 2, but 21 of those 27 pairs lie in assemblies below the contiguity bar, and above it 6 pairs cannot carry a test (Extended Data Fig. 7). We report it as unresolved rather than as a lineage in which the gene is decaying.

Discussion

Every vertebrate genome in a declared scope of 309 that is able to hold an IP_3 receptor gene on one contig carries all three paralogues, and no genome in the scope, contiguous or not, reaches the state that would license a loss. The count is zero on the family character and on each paralogue. That is an unusual result for a gene family that has existed since the origin of the vertebrates, since loss is the commonest fate of duplicated genes [19, 11] and a family with three members and five hundred million years of lineage splitting has had many opportunities to shed one.

The zero is worth stating because the search that produced it has a measured error rate, obtained from the same data. The false-negative rate, 140 of 923 cells overall, is not small, and a survey that reported zero losses without it would be reporting mostly the quality of bird assemblies. The rate falls to 5 of 563 cells above the contiguity bar, and an independent series that shares none of the state rules, the ryanodine receptor cell, falls to none of 189. Absences in this data are concentrated exactly where assemblies are poor, and where assemblies are good there are none. The contiguity bar that separates the two was fixed from gene geometry before any error rate was measured and survives calibration against both series; it is conservative against a five per cent target and about right against one per cent.

Three consequences for how gene-loss surveys are done follow directly from the measurements. First, an absence in a fragmented assembly is not evidence of loss, and in this scope the fragmented assemblies are not a random subset: the margin species, selected because their protein records looked incomplete, are very largely the same

genomes whose assemblies cannot hold the gene. Selecting genomes for apparent absence and then searching them selects for detection failure. Second, a paralogue-resolved absence is far more fragile than a family-level one, manufactured in 18 of 32 analytical settings against 2 of 32, because a fragment of one paralogue reads as evidence against another when the paralogues are 61 to 68 % identical. Third, the panel ablation says where a bait budget should go: paralogue coverage buys recall and phylogenetic breadth within the vertebrates buys almost none.

Retention of all three paralogues across the vertebrates is consistent with each carrying a function the others cannot fully replace. The mouse and human phenotypes of single-paralogue loss [7, 10, 5] and the exocrine failure of the double knockout [6] point the same way. Dosage balance, which is thought to retain duplicates from whole-genome duplications [20], is another explanation that this analysis cannot distinguish from subfunctionalisation; a zero loss count is compatible with both.

The result has limits that the data set. The scope is 309 assemblies, and nothing here extends to a lineage without one. Four cells in the two cyclostomes are unassignable rather than present, because the bait panel holds no labelled cyclostome *ITPR2* or *ITPR3*: the genomes carry three family loci apiece, which is the expected count, but which locus is which paralogue is not settled by this analysis. Paralogue labels throughout come from the bait panel's labelled sequences, and the family-level count, which needs no paralogue labels, is the more robust of the two. The one lesion signal, an *ITPR3* excess concentrated in birds, sits mostly in assemblies below the contiguity bar, where an indel excess cannot be separated from an assembly effect; the lineage is named and the mechanism is not. And the Markov rate models are refused rather than fitted, so the analysis states no rate of loss, only that none is observed.

The strongest form of the claim is therefore the one that names its own escape routes. A reader who wants a vertebrate IP₃ receptor loss from this data can obtain one: by refusing all reconstruction evidence and ignoring assembly contiguity together, which yields one genome at the family level, or by resolving paralogues in fragmented assemblies, which yields up to 45 loss edges. Both are descriptions of an analytical setting, not of a genome.

Methods

Scope

The denominator is 309 vertebrate genome assemblies, declared before the search and fixed. It is the union of two rules applied to NCBI assembly metadata [21]. The taxonomic rule takes one best assembly per vertebrate order, 161 orders, ranked by a stated function that prefers an annotated assembly, then RefSeq over GenBank, then assembly level, then scaffold N50. The margin rule adds 169 species computed from the protein census by four positive tests: the species' reference proteome returned no family record, carried fewer than three paralogues, held only records below the family's minimum length, or the species was an anchor needed for another reason. The same rank function resolves each margin species to an assembly, and a species selected by both rules is one row. The manifest records accession, taxonomy, assembly level, annotation source, contig and scaffold N50 and the rules that admitted each genome. The scope totals 552,535,418,825 bp. The deliverable unit is the genome × cell: three IP₃R paralogues plus the ryanodine receptor control cell in each genome, 1,236 cells, of which the 927 paralogue cells enter the loss analysis.

Bait panel and paralogue labels

The panel holds 38 baits, 30 IP₃ receptors and 8 ryanodine receptors, 121,294 residues, chosen by seven rules: a bait's paralogue label comes from the protein census rather than its own gene symbol; it is full-length with the complete domain architecture; there is one bait per paralogue per clade band, two in the bands that dominate the scope; every candidate passes a chimera screen against the IP₃R and ryanodine receptor profiles; the ryanodine

baits serve as a presence control; existing profile seeds are reused; and the panel is scoped to this sweep. Six slots have no labelled, full-length record and stay empty (*ITPR1* and *ITPR2* in cartilaginous fish, *ITPR2* in the coelacanth grade, all three in cyclostomes). Three unlabelled baits from those grades keep their genes inside the family without assigning a paralogue. The chimera screen is itself tested on every build with three constructed failures (an IP₃R–ryanodine fusion, a mis-joined model and a truncation), each of which must be rejected by the rule responsible.

Paralogue identity in this paper therefore rests on the labelled baits, and a cell is called for the paralogue whose bait scores highest at its locus, after the family has been settled by which family’s baits align at all. No cell call uses flanking-gene synteny or a gene tree. The family-level coding of the loss analysis uses no paralogue label at all.

Spliced alignment and rescue

Baits were aligned to each genome with miniprot [14]. Its maximum intron parameter decides whether a long gene is split, and a split IP₃R reads as a fragment, so the parameter was measured rather than taken as the default of 200 kb. Across an eleven-species Ensembl panel [22], the widest intron of any IP₃R gene is 152,216 bp (human *ITPR1*) and of any ryanodine receptor gene 227,927 bp (human *RYR2*). The sweep uses twice the widest measured intron scaled by genome size, floored at the aligner’s default and capped for tractability; the two genomes on which the cap binds are flagged. Cells left empty were re-asked with a whole-genome tblastn search [15], and each region was attributed first to a family and then to a paralogue against the full panel. The paralogue-attribution margin arrived from a sister project as 0.333; measured at 542 loci whose paralogue the assembly’s own annotation establishes, 368 of them fell below it, so the margin in force is 0.22, the highest value that rejects none of those calls. All 53 rescue regions that overlap an annotated gene of known paralogue were attributed in agreement with it.

The positive control

The ryanodine receptors are present in every vertebrate and are searched in the same run by the same aligner. A genome whose control cell is not found has an assembly or pipeline problem, and its IP₃R cells support no claim. The control failed in 0 of 309 genomes.

States

Each paralogue cell was restated by eight ordered rules, first rule to fire wins, each recording the number it fired on (Results). Coverage is aligned bait length over bait length, with the bar inherited from the sweep. The cyclostome rule reads the sweep’s own per-genome locus counts: a cell with no locus in a genome carrying family loci no cell claimed is paralogue-unassignable. The contiguity rule is the bar described below. The rule order is tested: moving the cyclostome rule after the absence rule must fail the suite.

Reconstruction across contigs

For each cell with no locus, each reference bait was taken in turn, never as a union, and its non-redundant coverage by translated-search hits outside every locus the aligner clustered anywhere in the genome (any bait, padded) was computed, with the fraction of reference residues covered once (tiling) reported beside the total (pile-up), and the genomic exonic length on the subject side as gene-equivalents. The bar was calibrated against 44 positive regions (cells with no other explanation), 9 negative regions (attributed to a paralogue already placed at a locus in the same genome) and the 20 co-shattered regions kept as their own population. Youden’s index [16] is reported with the gap’s two edges, and the operating point is the gap’s midpoint. The calibration refuses to return a threshold on too few regions or on unseparated populations.

Synteny reach

Each rescue region was offered to a flanking-gene paralogue caller with a four-key floor, whose accuracy was measured on its own leave-one-genome-out calibration and binned by the number of informative keys, so that reach and accuracy are read on one axis.

Contiguity bar and false-negative rate

The contiguity bar is 142,212 bp of contig N50, the median genomic span of 28 IP₃R genes measured across the Ensembl panel. It was fixed from gene geometry before any error rate was computed. Because no loss is reconstructed, the IP₃R control series is every cell the states call present, 923 cells (the four unassignable cells are excluded), and a false negative is a cell in that series the ledger did not grade as found. The ryanodine receptor series is the 309 control cells, which use none of the state rules. For each series we report the rate with a Wilson interval, a logistic fit of found against log₁₀ contig N50, a Mann–Whitney comparison of contig N50 between found and missed cells, a Fisher test of chromosome-level against lower assemblies and of one series against the other, and the residual rate at every candidate floor with the genomes each floor retains by clade. A constructed case requires a false negative to be reachable.

Panel ablation

Eighteen reduced panels were simulated beside the full one by dropping baits from the retained miniprot output and re-running clustering, the identity floor and the cell assignment in full. Scoring best coverage over all alignments instead would let one paralogue's bait stand in for another at the same gene, so the full chain is reproduced. The full-panel simulation must reproduce the committed ledger cell for cell, and it does in 1,236 of 1,236 cells. Every call change, gains included, is recorded with both baits and both coverages.

Tree, parsimony and rate models

The tree is NCBI taxonomy over all 309 assemblies, one tip per accession, unary nodes kept so that the edge a loss is placed on does not move. It is an input; its 49 polytomies are reported, not resolved. Losses were counted by Dollo parsimony [17] as the maximum (every loss edge) and the minimum (one per parent carrying any), which differ only under a polytomy. The family character's gain is pinned at the root and each paralogue's gain is left free. The sensitivity grid is the eight-rung evidence ladder crossed with the cyclostome and contiguity rules on and off, 32 settings, on the family and paralogue codings and under unit, ultrametric and calibrated branch lengths. Equal-rates, all-rates-different and irreversible Mk models [18] were profiled along a rate grid for every combination, and a fit is refused, with its reason, when the character does not vary.

Reading-frame integrity

Frameshifts and internal stops were counted per thousand aligned residues at every placed locus. The bar is the upper quantile of a population the screen never scores: full-coverage loci above the contiguity bar whose own annotation names them as family. Density was correlated with contig N50, bait identity and length. Each cell was compared with the same genome's other family loci in a paired sign test, ties dropped and counted, once unmatched and once identity-matched, and the tests were corrected together by the Benjamini–Hochberg procedure [23], then stratified by vertebrate class with the contiguity bar applied within each stratum. The verdict is one-sided: a locus without stops is not a pseudogene, and a handful of stops does not make one.

Negative controls and reproducibility

Every stage runs a suite of constructed negative controls before writing any table, each of which must be rejected by the rule responsible (for example, a reconstruction without the locus exclusion must reassemble a missing

gene out of its paralogues and fail). Everything downstream of the alignments is offline and deterministic, and every committed table carries a SHA-256 in its stage's statistics file.

Figure legends

Evidence for each paralog varies by vertebrate class

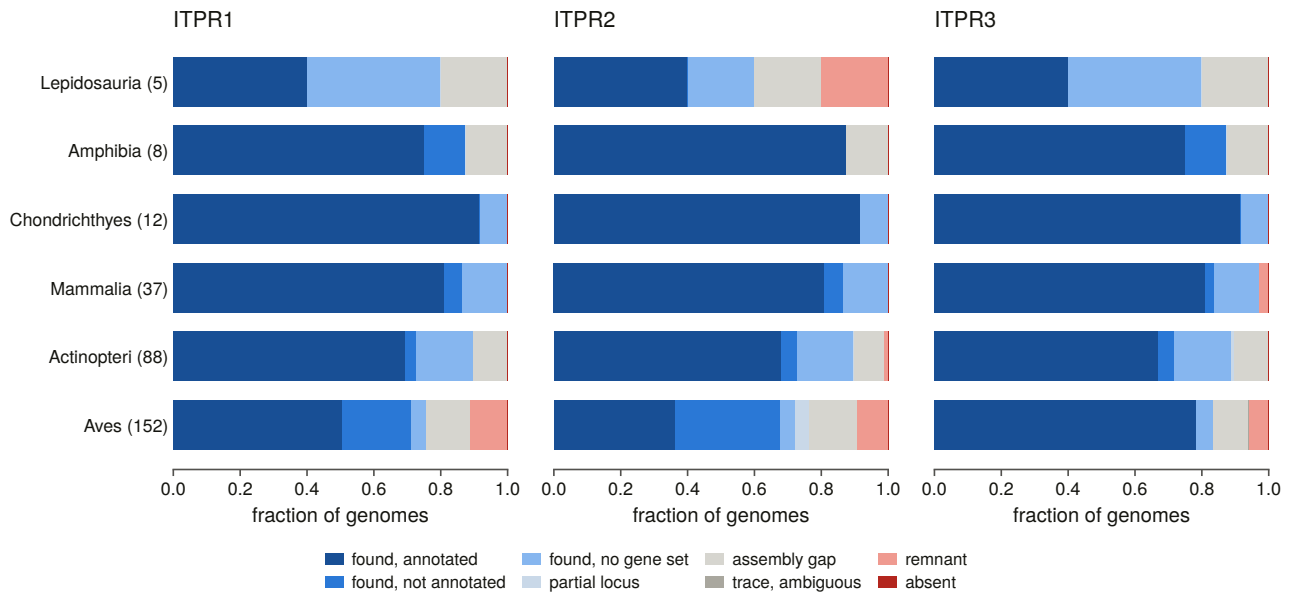


Fig. 1. The three IP_3 receptor paralogues are found in nearly every vertebrate genome searched. Each panel is one paralogue, each row a vertebrate class, and the bar stacks the evidence for the gene as a fraction of that class's genomes, from a full-coverage locus through truncated and fragmentary evidence to the absent call. The red that remains sits in a few classes, and those are the classes with the poorest assemblies, so this raw ledger is the problem the later figures resolve rather than a map of loss.

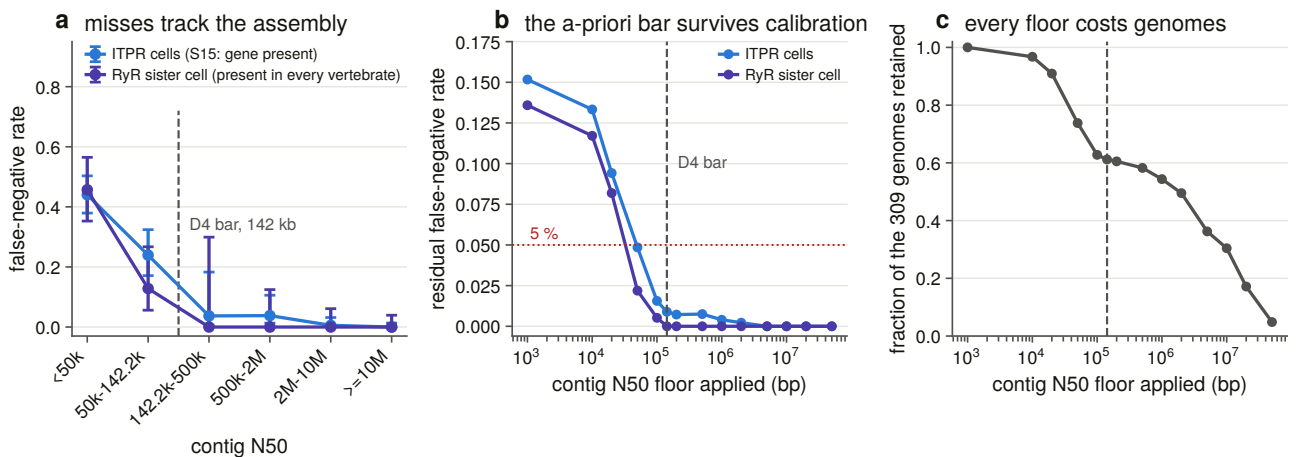


Fig. 2. The search's miss rate falls with assembly contiguity in two independent control series. (a) The share of known-present cells the ledger missed is plotted against contig N50, for the IP_3 receptor series and for the ryanodine receptor series, which uses none of the state rules. (b) The residual miss rate is scored at every candidate contiguity floor, with the a-priori bar marked. (c) Each floor removes genomes, and the panel counts them. Agreement between the two series is what stops the rate being read as the search agreeing with itself.

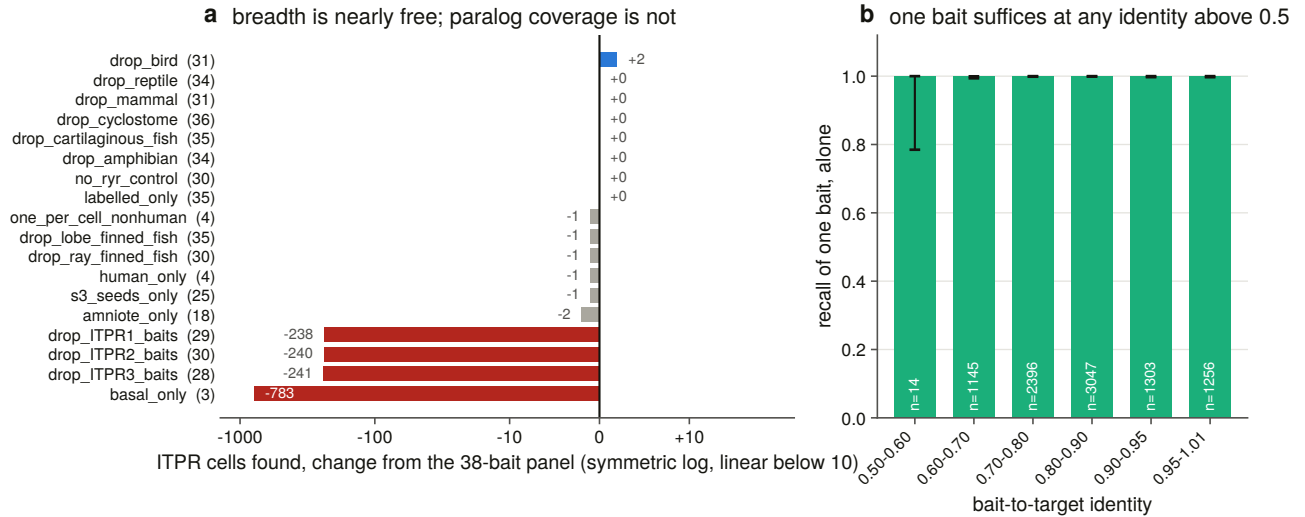


Fig. 3. Parologue coverage decides the panel's recall and phylogenetic breadth barely matters. (a) Each of 18 ablated panels is drawn as its change in recovered cells from the full 38-bait panel on a symmetric-log axis; only the ablations that remove a parologue's own baits, and the one that removes every labelled bait, move far. (b) A single bait recovers its cell at any identity to the gene above 0.5. The simulation behind both panels reproduces the committed ledger cell for cell.

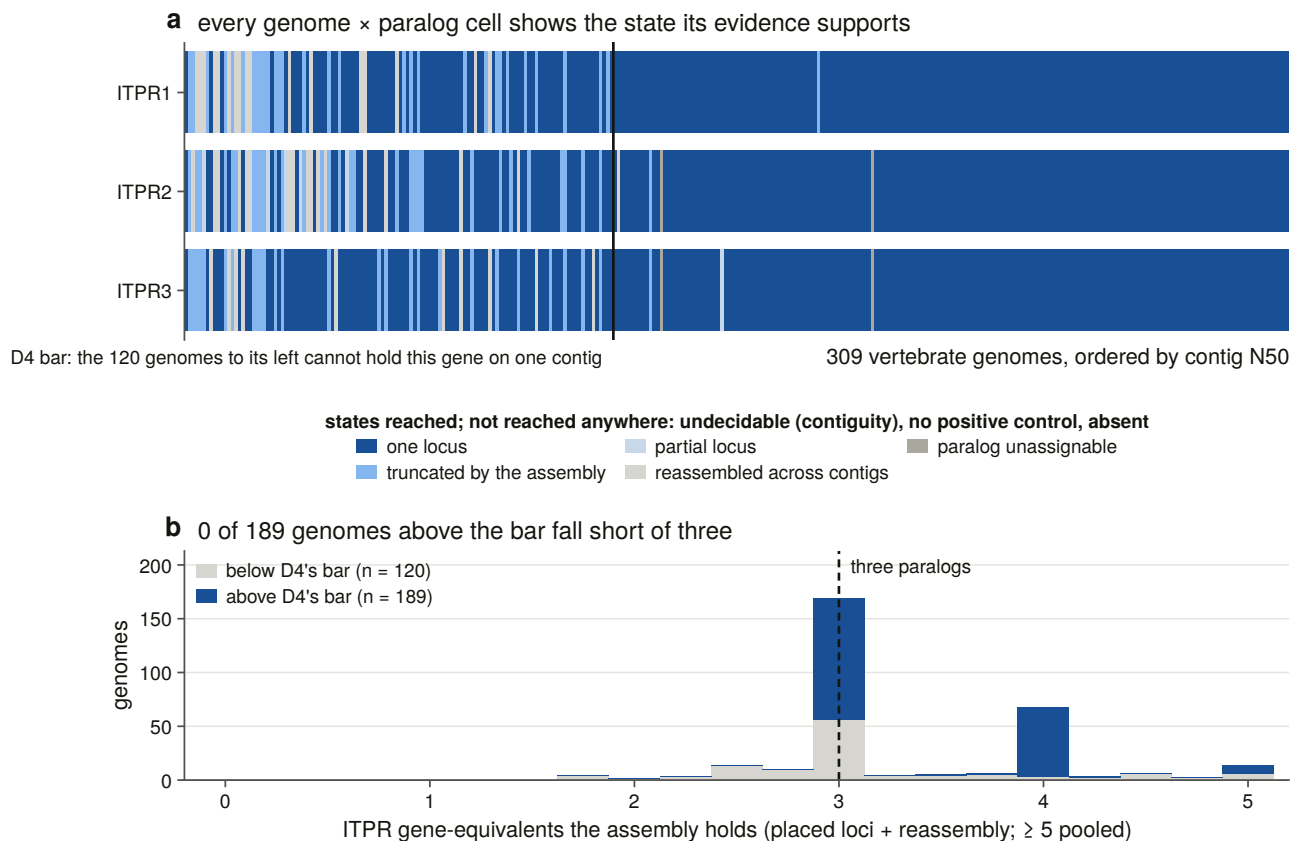


Fig. 4. No genome $\times$ parologue cell is absent once every alternative explanation is tested. (a) All 927 cells are drawn in the state their evidence supports, with genomes ordered by contig N50 and the contiguity bar marked. (b) Summing genomic sequence per genome gives the gene-equivalents each assembly holds; above the bar every genome holds three. The absences of the raw ledger have become truncations, reassemblies and four unassignable cyclostome cells.

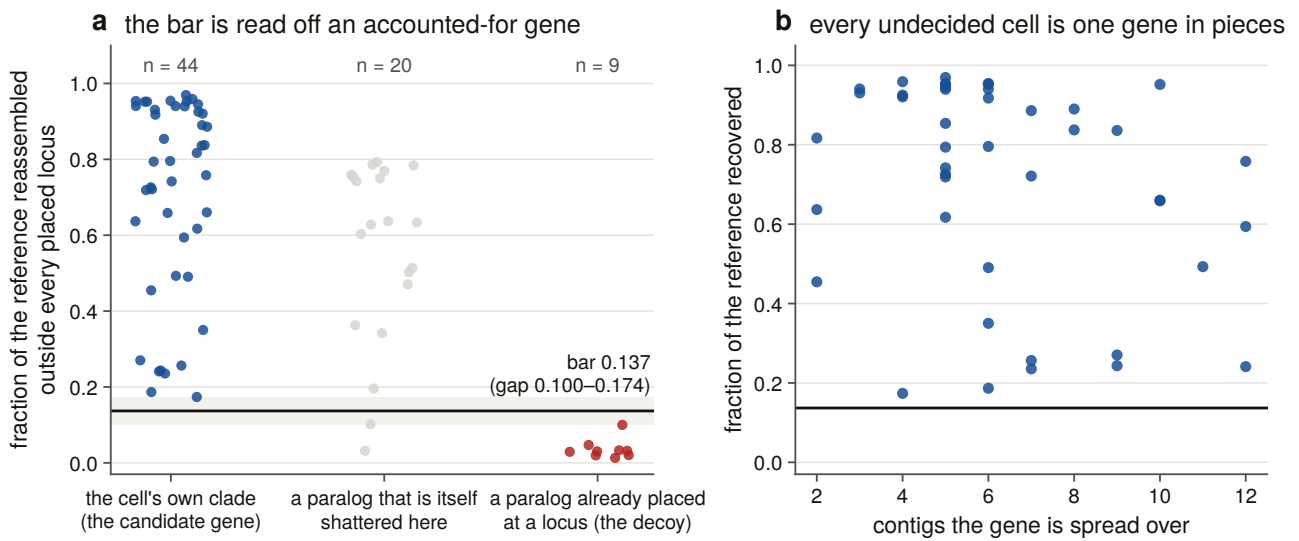


Fig. 5. A calibrated bar separates a shattered gene from cross-paralogue similarity. (a) Reassembled coverage is shown for the candidate regions, for the decoys attributed to a paralogue already placed elsewhere in the same genome, and for the co-shattered regions between them, with the gap shaded and the operating point drawn. (b) Every undecided cell clears the bar however many contigs its gene is spread over. The third population was not anticipated, and separating it is what made the calibration separate at all.

Dollo losses manufactured by each setting of the rule chain (309 vertebrate genomes); the violet ring is S15a's own operating point

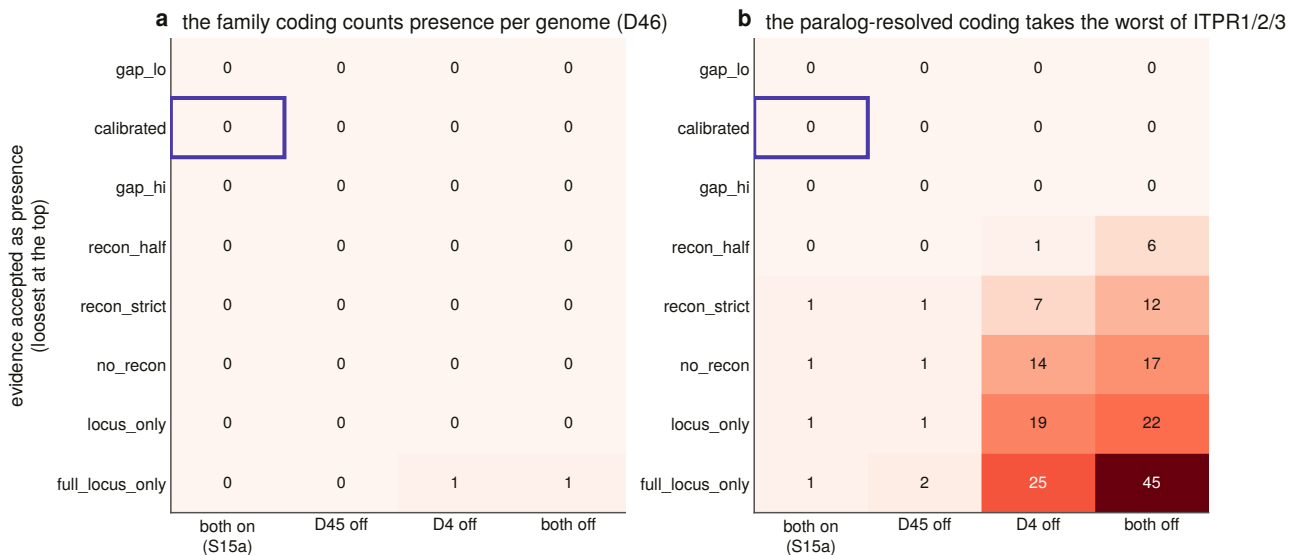
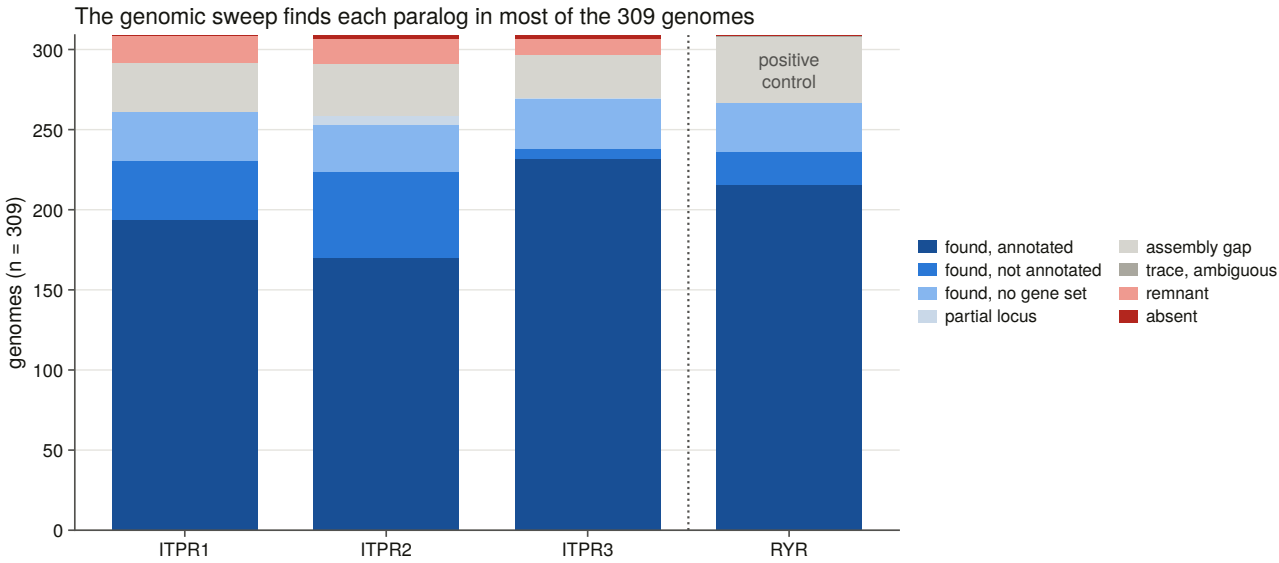
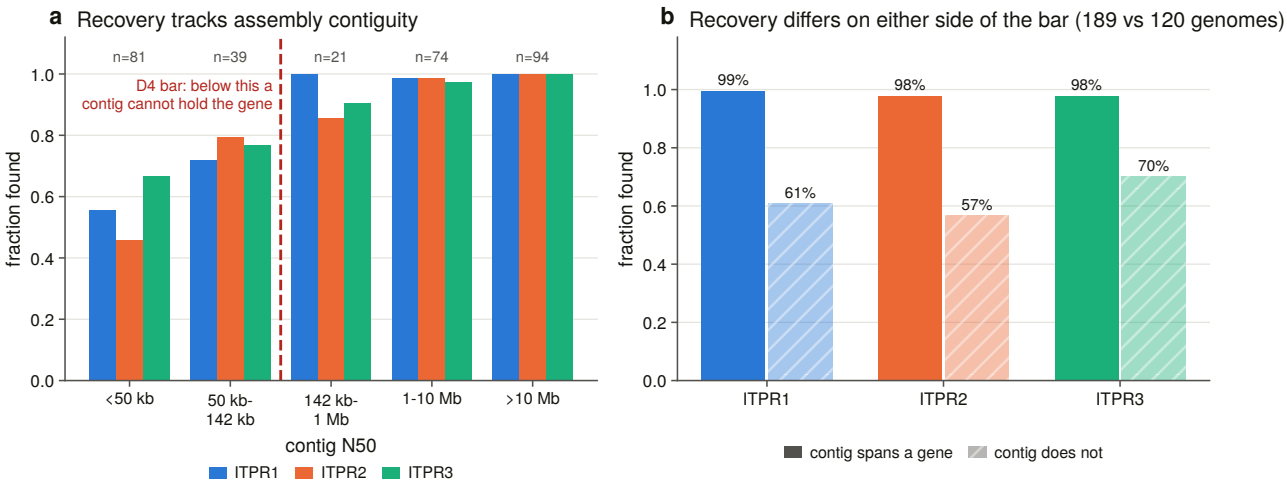


Fig. 6. A family-level loss is almost impossible to manufacture from this data and a paralogue-level one is easy. (a) The Dollo loss count is given for the family-level coding in every setting of the evidence ladder crossed with the cyclostome and contiguity rules. (b) The same grid is shown for the paralogue-resolved coding. The operating point is ringed in both, and a zero is printed as a zero because an empty cell would read as a setting nobody measured.

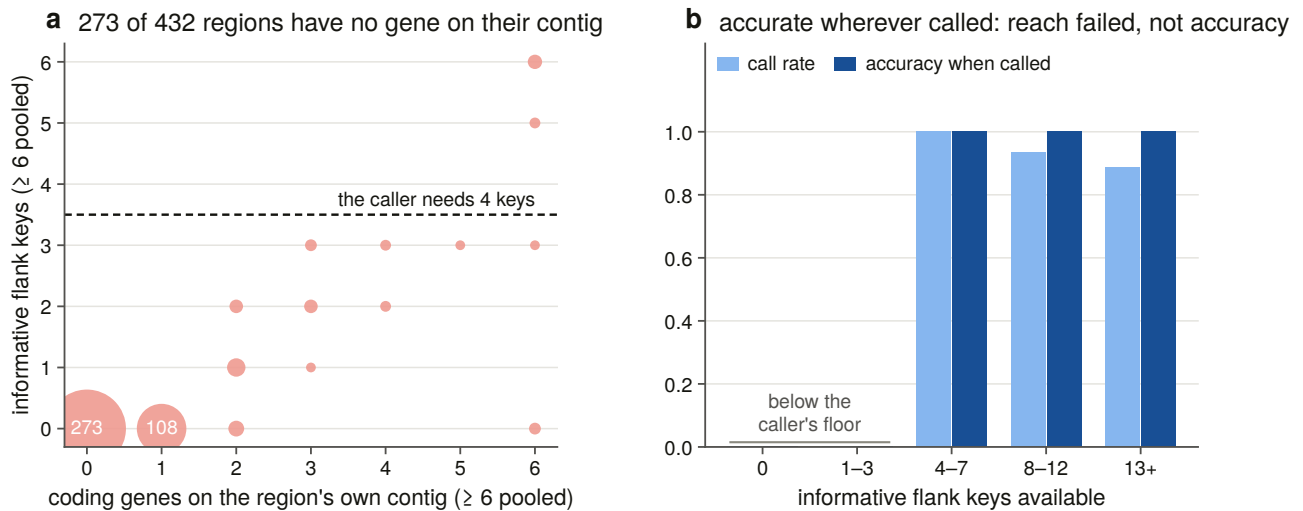
Extended Data



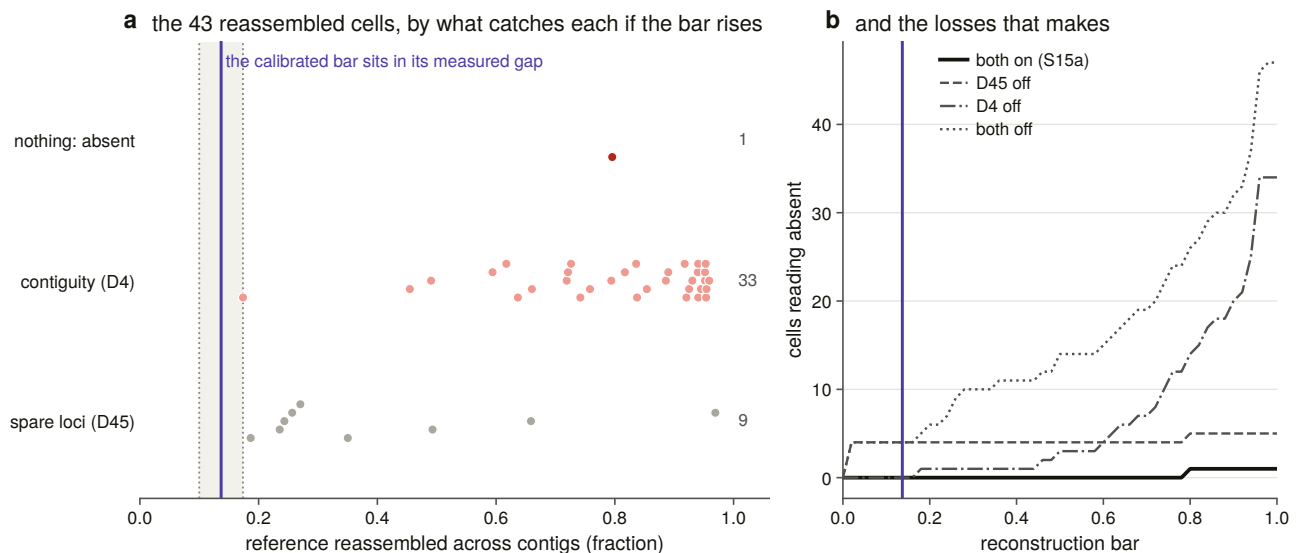
Extended Data Fig. 1. Each of the 927 genome $\times$ paralogue cells carries one ledger status, stacked per paralogue from the strongest evidence to the weakest. Four cells reach the absent status, and all four are in the two cyclostome genomes. They remain the strongest negative the search itself produces, and the state rules show that they are paralogue labels the panel cannot resolve.



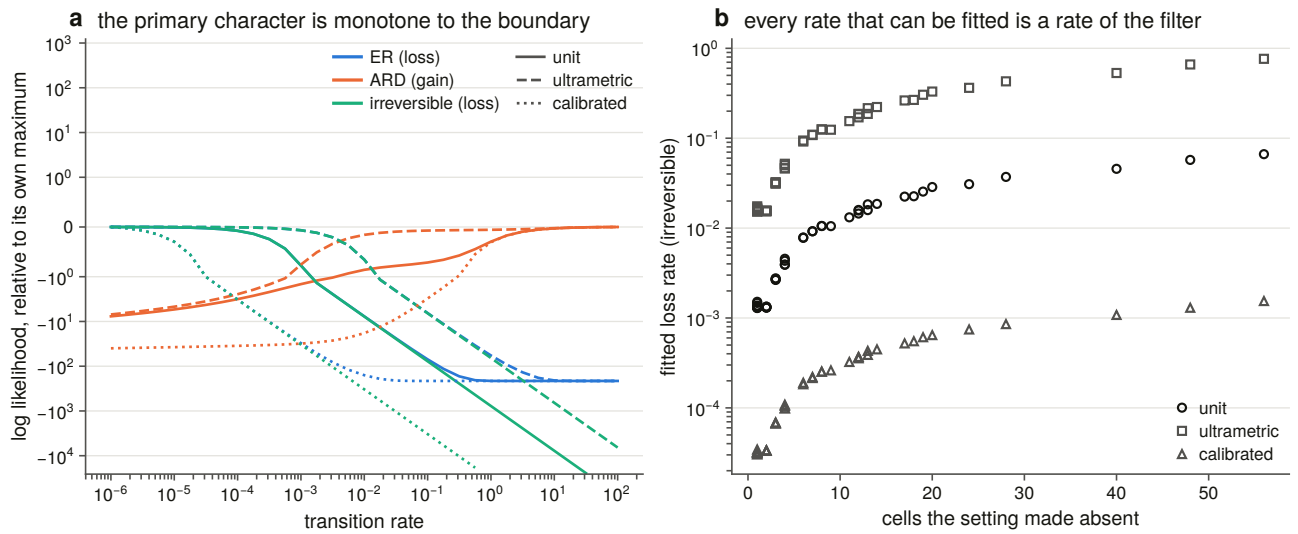
Extended Data Fig. 2. Recovery rises with assembly contiguity, and the contiguity bar splits the scope into two populations. (a) The fraction of cells found is plotted in bins of contig N50 that are placed on the bar rather than across it. (b) The same recovery is collapsed to the 189 genomes above the bar and the 120 below it. A sliding window across the bar would report a recovery rate that no genome in the window has.



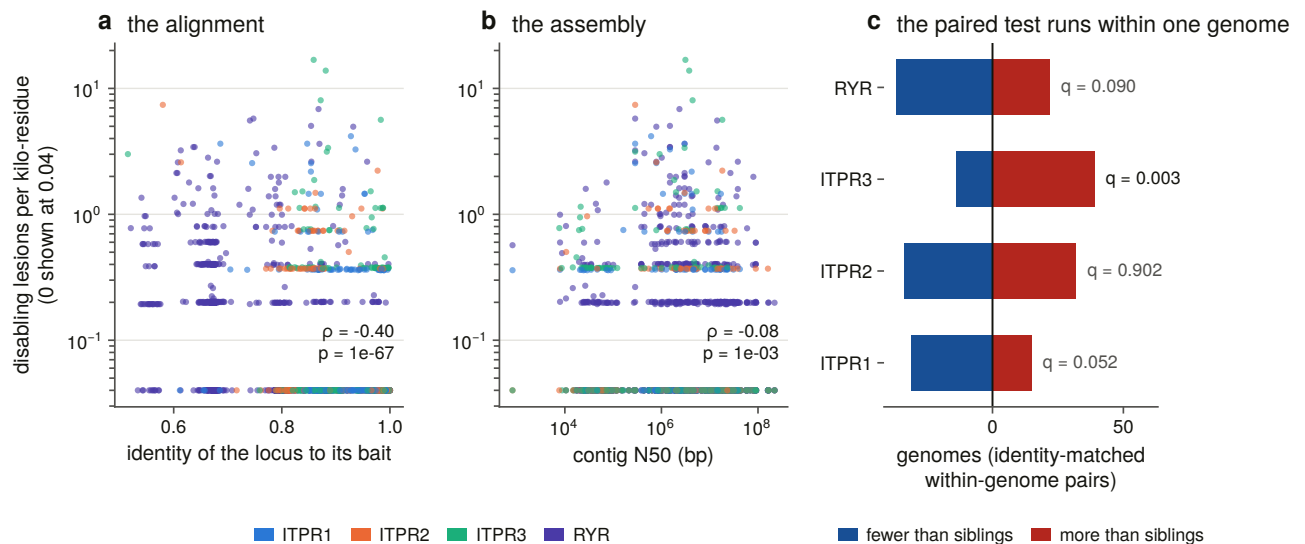
Extended Data Fig. 3. Synteny cannot place the fragments because their contigs carry nothing to read. (a) Each rescue region is placed by the number of coding genes and informative flanking symbols on its own contig, with the caller's four-key floor drawn. (b) The caller's call rate and accuracy are plotted on the same key axis. The caller is right whenever it calls, so what fails here is reach.



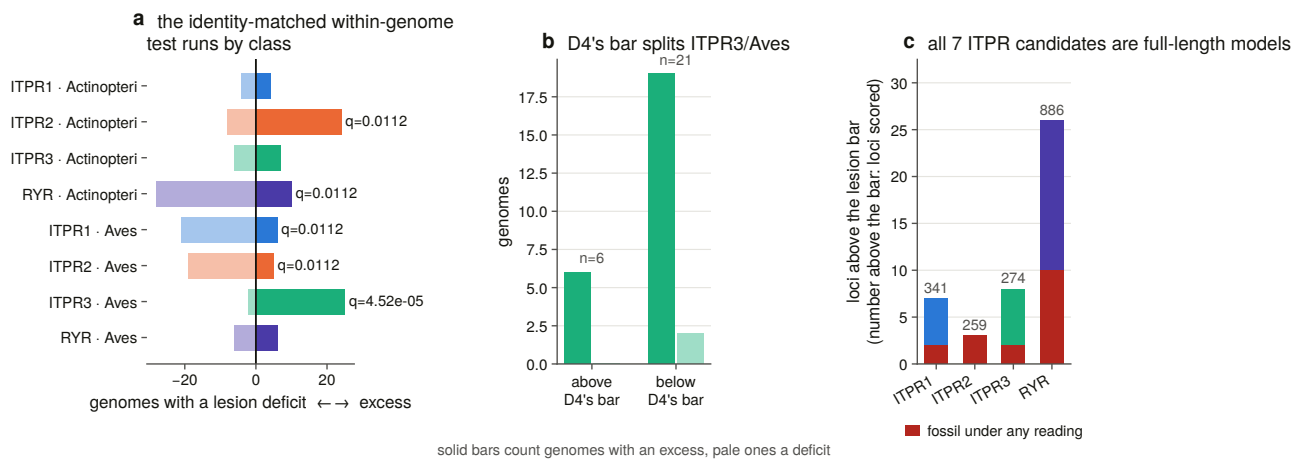
Extended Data Fig. 4. Moving the reconstruction bar across its own measured uncertainty changes no cell. (a) Each of the 43 reassembled cells is placed against the bar and its gap, on the row of the rule that would catch it if the bar rose; the within-row offset is a point's rank, so the drawing is reproducible. (b) The count of absent cells stays at zero with both protective rules on until the bar is raised far beyond the gap.



Extended Data Fig. 5. The rate models are refused because the character never changes, and the refusal is measured. (a) The likelihood is profiled along a rate grid for every model, axis and branch-length scheme, and every curve runs to the edge of the grid. (b) Every rate that can be fitted in the sensitivity grid rises with the number of cells its setting made absent, so it describes the filter rather than the family.



Extended Data Fig. 6. Lesion density follows bait identity far more than assembly contiguity. (a) Frameshifts and stops per thousand aligned residues are plotted against identity to the bait. (b) The same density is plotted against contig N50. (c) The paired within-genome test compares each cell with its own genome's other family loci, which removes the assembly. Density is logarithmic with a zero band, because most intact loci carry no lesion.



Extended Data Fig. 7. The one surviving lesion signal is a bird result that the contiguity control leaves unresolved. (a) The identity-matched sign test is run within each vertebrate class. (b) The bird *ITPR3* pairs are split at the contiguity bar. (c) The loci above the lesion bar that any fossil reading flags are counted, and all are full-length models. Most of the bird evidence lies below the bar, where an indel excess cannot be told apart from an assembly effect.

Data availability

Every table and figure this paper reports is committed under `results/` (`genome_manifest.tsv`, `s5_baits/`, `genome_ledger/`, `loss_dynamics/`, `loss_counts/`, and the contiguity and panel halves of `methods/`), with a SHA-256 for each file in `papers/retention/deposit_manifest.tsv`. Genome assemblies are retrievable from NCBI by the accessions in `results/genome_manifest.tsv`. Per-genome sweep output (miniprot GFF, rescue output and a summary per genome) is regenerated by `python scripts/s5_run_sweep.py` and is listed, with the command for each bulk class, in `papers/retention/deposit_notes.md`.

Code availability

The analysis scripts are in `scripts/`. `python scripts/s26_assemble.py --paper retention` rebuilds this paper from the committed tables, re-checks every load-bearing number against its source table, and exits non-zero if any has moved.

Author contributions

G.D. conceived and directed the project and set its questions and standards. The analysis code, the analyses and the text were written by Claude (Anthropic) under his direction; Chapter 16 of the accompanying thesis describes how the work was carried out.

Competing interests

The author declares no competing interests.

Companion papers

2. George Dickinson. The three vertebrate IP₃ receptors arose in two separate early vertebrate duplications. *In preparation* (paper 2 of 6 in this series; source in `papers/origin/`).

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Citations are rendered from `results/s0_baseline/references.tsv`, numbered in order of first appearance.

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